

x

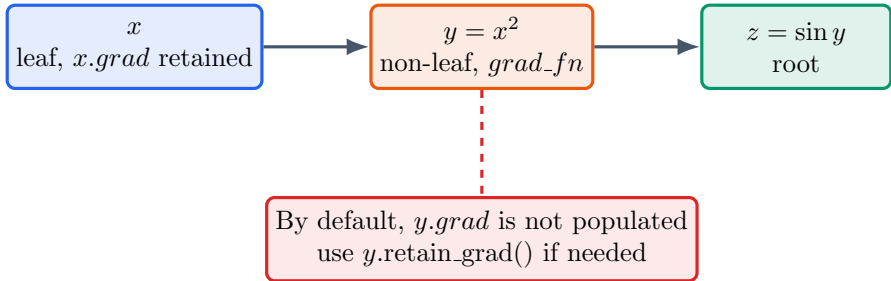
$$y = x^2$$

local: $2x$

$$z = \sin y$$

local: $\cos y$

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx} = \cos(y) 2x = 2x \cos(x^2)$$



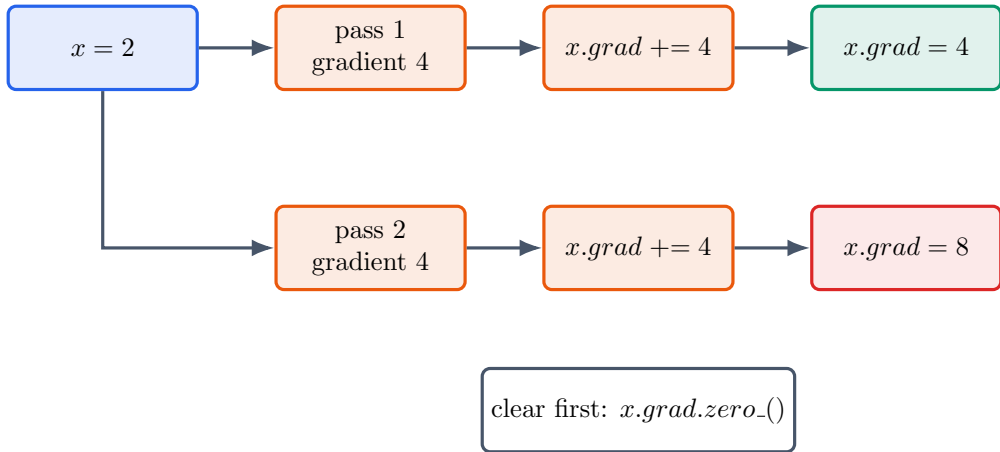
$$\mathbf{x} = [1, 2, 3]$$

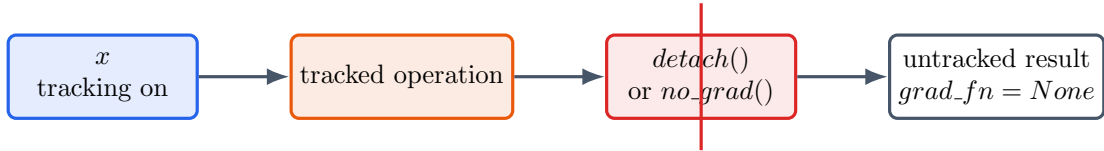
elementwise square
 $[1, 4, 9]$

mean
divide sum by 3

$$y = 14/3$$

$$\nabla_{\mathbf{x}} y = \left[\frac{2}{3}, \frac{4}{3}, 2 \right]$$





gradient path stops here

L.backward() : $\partial L / \partial w, \partial L / \partial b$

forward pass

